

Key Punch Data:				County Number	Site Number	Emission Point #	Action Ty.	Date	Agency/Indiv. ID
								Mo Day Yr	
Li	1	1011101	1212201	101010131	10101	1919171	11121	101918131	10101
		1	3	4	7	8	12	13	15
						16	18	52	54
								59	66
									68
									80

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

$$\begin{array}{r} 1010151 \\ 16 \quad 18 \end{array}$$
$$\frac{11}{78}$$

1 C
8 0

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Company: Martin Marietta Cement
Site Location: P.O. Box 529, Lyons 80540
Contact Person: Gerald Anderson
Telephone No.: 534 - 4206

Inspected by: G. Mathews S. Martino.

County: Boulder

Time: 9:30 am

EIS Update: Included

16.0 Hrs.* Permit No. See individ.
print.

Inspection: Routine ☒ Complaint ☐ FA ☐ 16.0 Hrs.
Emission Points Subject to NSPS: Kiln and clinker coolers

1. Annual inspection per State contract.

1. Annual inspection per state contract.
2. Inspection occurred over a 2 day period : 9/1/83, 9:30 am - 12:45 pm, and 9/7/83, 2:00 - 4:30 pm. Contact persons during these visits were Gerald Anderson, Manager of Operations; Roy Cook, Process Manager; Sam Angelo, Production Supervisor; Marvin Berg.

3. EMS Point #1. Kiln Stack. Kiln, spray tower, and combustion chamber vent through the "kiln baghouse," while the kiln alone vents through the "alkali baghouse." These 2 baghouses then vent through the kiln stack. No problems were mentioned with the alkali baghouse maintenance. However, Martin-Marietta has been having trouble with the kiln baghouse. Collected dust has been accumulating in the hopper with subsequent disposal difficulties. This reduces production. A solution has not been selected. A EPA RM 9 opacity observation was taken. Average opacity was 2.6%. See attached report.

1982 Production data: Tons of cement produced: 454,296 tons.

Tons of coal burned: 85,308 tons

Coal is 0.56% - 0.71% S.

No. 6 fuel oil burned: 256,003 gal

Natural gas burned: 65,580 MCF

COLORADO DEPARTMENT OF HEALTH
AIR POLLUTION CONTROL DIVISION

ANALYTICAL REPORT OF VISIBLE EMISSION ~~VIOLATION~~ OBSERVATION

COMPANY Martin Marietta Cement Plant DATE 9/1/83

ADDRESS OF SOURCE Lynx Colorado
City Street County AQCR

STACK NAME Kiln Stack

BACKGROUND SKY CONDITION blue

RELATIVE HUMIDITY unknown % PSYC. — WEATHER BUREAU — AIR TEMP. 80 ^{est} OF

WIND SPEED calm or south 0-5 WIND DIRECTION (From) south

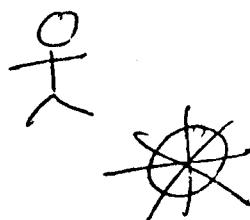
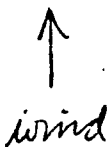
PLUME CHARACTERISTICS (Color, etc.) —

STACK HEIGHT est 120 Feet OBSERVER'S LOCATION 120 Feet SE of Stack

WET PLUME or DRY PLUME: (If WET, estimate the breakpoint distance from the stack)

CLOCK TIME	0	15	30	45
10:20	0	5	5	5
10:21	01	5	5	0
10:22	02	0	5	5
10:23	03	5	5	0
10:24	04	0	0	0
10:25	05	0	0	0
06				
07				
08				
09				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				
25				
26				
27				
28				
29				

SKETCH: Include 1) Wind Direction; 2) Sun Position; 3) Observer Position; 4) Approximate Location of stack; 5) North Arrow.
If more than one stack is located at the plant site specifically identify the stack being read. Include remarks.



2.6%
24/65.
50
150

OBSERVER: Geo Mathews DATE 9/1/83
PERSONS PRESENT DURING THE EVALUATION:

Name: Geo Mathews
Title: Coordinator

SIGNATURES: Duplicate of this Analytical Report Given To

Name:
Title:
Date:
Time:
Signature:

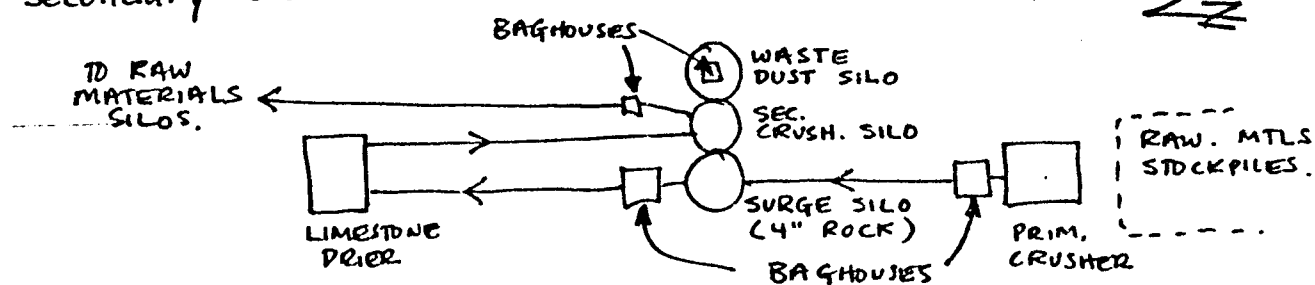
Martin Marietta Cement, Source No. 3
pt. 1, 1983 Inspection report continued

ØØ1 Kiln Stack continued

In addition to coal, natural gas, and residual fuel oil, Martin-Marietta has been using some waste oil. Supplier is Milt Adams. Because of potentially hazardous materials, including lead, in waste oil and because this fuel was not listed on original APEN, APCD is interested that company file a revised APEN. As a matter of fact, coal is the only fuel listed on APEN.

4. EMS Point ØØ2. Raw materials crushers. These points were inspected on 9/7/83. Material is hauled from the shale quarries and piled in area south of primary crusher. An estimated 150,000-200,000 tons of raw material are normally accumulated. There are four baghouses associated with this part of the process:

1. primary crusher baghouse.
2. surge silo baghouse.
3. baghouse at a conveyor drop point on conveyor from secondary crusher silo to the raw material silo.
4. baghouse on the waste dust silo which collects for both the secondary crusher and waste dust silo.



The primary crusher has a design rate of 400 tons/hr., MMC is now feeding ~200 tons/hr. The secondary crusher is designed and now operated at 150 tons/hr. During 1982, 746,396 tons of raw material were processed.

Martin Marietta Cement, Source No. 3
pt. 1, 1983 Inspection report cont.

φφ2 Raw materials Crushers continued:

Because the waste dust silo baghouse serves also the secondary crusher and silo, it seems logical to put the waste dust silo under EMS Point φφ2. The hot waste dust is mixed with water and loaded into dump trucks which haul to quarries. Because the waste dust is a calcined material, too much dust can cause explosions. Hot dust can cause steam pockets which explode. At the time of inspection on 9/7/83, quantities of steam were emitted from loading operation. Company personnel acknowledged that this is a problem point when workers try to load from an empty silo. It can be controlled by good management.

5. EMS Point φφ3. Limestone dryer.

Dryer was not operating during either inspection visit. 9/1/83 unit was down for routine weekly mtnc. 9/7/83 unit was down because all 960 bags were blinded. Evidently the dryer dropped below the dew point.

The final permit C-12,444-1 has not been issued for this point. Stack testing was done in Sept. 1982 and submitted to APCD in Nov. 1982. APCD engineer John Plog will not issue permit for the following reasons:

1. Test was performed on natural gas and Mr. Plog says it should have been done on fuel oil. Permit can be issued on gas use only stipulation, but Mr. Gerald Anderson indicated that MMC does want the fuel flexibility and may use coal in the future.

2. No fuel consumption or product rate data for the test runs were provided.

3. The emission reductions required by the letter dated Oct. 18, 1979 attached to permit need to be confirmed.

During 1982, 167,487 MCF of natural gas were burned in the dryer; no fuel oil was used.

6. EMS Point #4. A-frame and clinker storage.

Emissions out of the north and south doors of the A-frame can be significant. No opacity readings were taken at times of inspection visits, and no problems were noted at this time. However, past observations note that the potential for significant visible emissions from doors exists. The Oct. 18, 1979 letter attached to the C-12,444 permits indicated that there will be no emission points from the A-frame. MMC will be able in the future to keep one door closed and provide a dust curtain on the other access door.

Large quantities of clinker are currently stored outside the A-frame. This clinker was manufactured in Utah and transported to Lyons. A permit for storing outside needs to be obtained.

On 9/1/83 a small conveyor and screen was observed outside the south side of the A-frame. It was not observed working so no opacity readings were taken. This equipment is used for loading Utah clinker into the A-frame. The need for a permit and to keep emissions < 20% was brought to the attention of MMC. On 9/7/83 the conveyor + screen had been removed.

The Utah clinker was originally 35,000 tons; now it is about 25,000 tons.

7. EMS Point #4 Site fugitive dust.

No problem noted. MMC is continuing to use an 18% lignin sulfonate solution on the haul roads. This winter MMC might build a 6" andesite road base. Vehicles travel 75-100 miles/day; 5 days/wk.

8. EMS Point #6. Coal handling

Coal is now all delivered by truck. A receiving building northwest of plant takes coal into below grade hopper. A canopied conveyor transport coal to the pile building on the top of which is located the coal crusher. At ground level in the pile is the coal mill. On the top of the pile is a baghouse

Martin Marietta Cement, Source No. 3
4.1, 1983 Inspection Report Continued.

EMS Point #6. Coal Handling, continued.

but a notation on the 1982 inspection report indicates that all coal particulate emissions are circulated back into system. This should be confirmed by MMC. In 1982, 85,308 tons of coal were burned in plant.

9. EMS Point #7. Quarries. Inspected 9/1/83.

Martin Marietta has acquired a new Ingersoll Rand DM25 drill rig. Primary dust control is accomplished by a cyclone-mechanical filter; back-up secondary control would be provided by a water injection system. Fifty gal. storage capacity is available on rig. We observed test drilling into shale at the E-Pit site, south of Hygiene Road. About 5% emissions were observed during pulses of the cyclone at the collector vent. Particulates are dropped to ground by means of a tube outfitted with a dust skirt. Emissions can be as high as 40% when dust escapes through slits in the skirt. These slits should be closed. The actual drilling location is also fitted with a skirt, called a "drilling shroud," at which point emissions at less than 20%. An exception is during "blow-out," an infrequent procedure, when drilling is interrupted and resumed. This causes high emissions.

D-Pit is actively being mined at this time. Emissions estimated to be 40% - 60% were observed during the truck loading process from the working face down in the pit. MMC should submit a dust control plan for existing and future mining activities.

The silica quarry from which MMC obtains sand on the west side of U.S. Highway 36 was inspected for the first time. Apparently Martin-Marietta owns the land, but the operation is owned

Martin Marietta Cement, Source No. 3
September 1, 1983 Inspection Report continued

EMS Point #7 continued

and operated by Ron and Leonard Loukonen. The six (6) acre site has a drill rig, primary jaw crusher, primary screen, secondary crusher. Power is provided by a generator. The Loukonen brothers need to be contacted for permitting requirements.

10. During the two tours of the plant several emission points were noted that have apparently been overlooked during previous inspections. These are discussed below and should be entered into the EMS/EIS Systems.

A. Clinker coolers. The clinker cooler vents 90% of its exhaust into two baghouses, located on the east and west sides of the kiln. 10% of the clinker cooler air goes to the coal mill and drier which then goes to the combustion chamber of the kiln. At the time of the 9/1/83 inspection, the east clinker cooler baghouse was out of service for regular 1-day maintenance. The west baghouse was carrying the entire load. The normal operating temperature of the baghouse is 300°-350°F. When the clinker coming out of the kiln is too hot, as it was on 9/1/83, the clinker is dropped into a holding shed next to the cooler. The west end of the shed was open and 40% - 50% opacity emissions were observed.

Also 15-20% opacity emissions were observed at the stack of the west clinker cooler baghouse. An official observation was not taken because of sun location and equipment interference.

On 9/7/83, 80 bags were replaced since the 9/1/83 inspection in the west baghouse. Emissions were 5-10%.

Mr. Anderson explained that the Nomex bags (400°F) now in use will be replaced by higher temperature Gortex bags (500°F) in order to accommodate the 500°-700°F clinker. Installation is expected in 6 weeks.

Martin Marietta Cement, Source No. 3
September 1, 1983 Inspection Report continued.

B. Homogenizing Silo. Baghouse located on top of silo. 0% opacity observed.

C. Finished Products Silos. Silos have 5 emission points: three (3) baghouses on the top of the silos for the silos themselves; one (1) baghouse venting to the side of the southwest corner of the building for the individual packing operation; one (1) vent point for two (2) baghouses collecting from the loading operation.

0% opacity observed at all points.

D. Mill Building. Four (4) rectangular ducts^{are} located along the north side of the second story of the building. The two (2) on the east side collect from the raw mill (#2 from east) and auxiliary equipment (#1 from east); the two on the west side collect from the finish mill. 0% opacity observed at time of inspection, but high emissions observed at other times. Emissions from raw mill are calculated w/ limestone dryer (SCC Code 3-05-006-13); but clinker grinding needs to be calculated (3-05-006-17). We were told that equipment is controlled by baghouses.

E. Silica and iron ore silos. Hopper, chutes, and elevator emissions are collected by a baghouse, which operates only during the unloading process into the silos.

II. Status: Not in compliance. See attached letter.

Time: Travel : 0.75 hrs.
Inspection: 6.25 hrs.
Write-up: 9.00 hrs
16.00 hrs

Martin Marietta

COMPLIANCE DETERMINATION WORK SHEET

Source p3 9/1/83

Emission Point ØØ1 Type of Source Kiln StackPermit No. C-12,444 (2) Is a Revised APEN Required? No

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
SO _x	continued		<u>Fuel Oil</u> 256,003 gal/yr. x (108) (1% Sulfur) x ton SO _x / 2000 lb = 14 tons <u>Natural Gas</u> 0 tons Total = 3069 tons/yr 3069 tons x (1 - 0.75) for control by SO ₂ adsorption to cement = 774 tons/yr.	3-90-004-02 3-90-006-02 AP42 8.6-4	YES

Emission Point ØØ1 Type of Source Kiln StackPermit No. C-12,444 (2) Is a Revised APEN Required? No.

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
NO _x	None specified	None	454,296 tons/yr x 2.6 lb NO _x / ton x ton / 2000 lb = 591 tons/yr	SCC Code Book, p. 3.7. 0-65. 3-05-006-6	
VOC	None specified	None	0	3-05-006-6	
CO	None specified	None	0	3-05-006-6	

*If No. Indicate what action has been taken.

Emission Point φφ2 Type of Source Raw materials c hars

Permit No. Grandfathered Is a Revised APEN Required? No

arks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	20% Opacity and No-Off Property Guidelines	Reg. No. 1 III.D.2.c.	0%	Observation	YES
	<u>RAW MATERIAL UNLOADING</u>	None	746,396 tons/yr x 0.2 lbpt/ton x ton/2000 lb. = 75 tons/yr.	SCC Code Book. p. 3.7.0-65 3-05-006-07	
	<u>RAW MATERIALS PILES</u>	None	746,396 tons/yr x 4.0 lbpt/ton x ton/2000 lb = 1493 tons/yr.	3-05-006-08	
	<u>PRIMARY CRUSHING</u> Design rate on 4/18/79 APEN: 200 TPH PE = 17.3 (200) ^{0.16} = 40.38 lb/hr.	Reg. No. 1, III.C.1.b.	746,396 tons/yr x 0.5 lbpt/ton x (1-0.999) baghouse x ton/2000 lb	3-05-006-09	

Emission Point _____ Type of Source _____
Permit No. _____ Is a Revised APEN Required? _____
Remarks: _____

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
	↓ 40.38 lb/hr x 7896 hrs/yr x ton/2000 lb = 159 tons/yr.		↓ = 0.2 tons/yr.		YES
	<u>RAW MATERIAL TRANSFER</u>	None.	746,396 tons/yr x 0.3 lbpt/ton x (1-0.99) totally enclosed w/ neg. pressure to a bagfilter x ton/2000 lb = 1.1 tons/yr.	3-05-006-12 end. APCD Fug. Dust Memo 9-30-81	

Martin Marietta Cement Co

EMISSION DETERMINATION WORK SHEET Source No. 3 9/1/83

Emission Point φφ3 Type of Source Limestone dryerPermit No. C-12,444 (1) Is a Revised APEN Required? No

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	144 tons/yr.	C-12,444(1)	$454,296 \text{ tons/yr}$ $\times 64 \text{ lb/t/ton}$ $\times (1 - 0.999) \text{ baghouse}$ $\times \text{ton}/2000 \text{ lb}$ $= 14.5 \text{ tons/yr}$	SCC code Book P. 3.7.0 -65 305-006-13	yes

Emission Point φφ4 Type of Source A-frame clinker StoragePermit No. None Is a Revised APEN Required? No

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	20 %	Reg. No. 1 II. A. 1.	< 20 % (est. 5 %)	Observation	yes

Martin Marietta Cement CON INCE DETERMINATION WORK SHEET Source No. 3 9/1/83

Emission Point 005 Type of Source Site fugitive dust

Permit No. None Is a Revised APEN Required? No.

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	No off-property transport guideline	Reg. No. 1, III. D. 2. e. f.	No off-property transport. Vehicles 90 miles/day x 250 days/yr x 3.0 lbPt/mi. x (1-0.85) cml. suppress. x 1 ton/2000 lb. = [5 tons/yr]	Direct observation SCC Code Book P. 3.7.0-28 3-05-020-11 and APCD Fug. Dust Memo.	yes

Emission Point 006 Type of Source Coal Handling

Permit No. Grandfathered Is a Revised APEN Required? No.

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	Unknown		No emission factor has been identified for coal crushing. mmc needs to verify that all emissions are returned to pile.		

Martin Marietta Cement

EMISSION DETERMINATION WORK SHEET

Source No. 3, 9/1/83

Emission Point 007 Type of Source QuarriesPermit No. Grandfathered Is a Revised APEN Required? No

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
PT.	20% and No. off property transport	Reg. No. 1, III. D. 2. d.	40-60%	Direct observation	
			Blasting 746,396 tons/yr. x 0.16 lb PT/ton x ton/2000 lb = <u>60 tons/yr</u>	Source needs to submit control plan. SCC Code Book P. 3.7.0-28 3-05-020-09	

Emission Point unassigned ⁰⁰⁸ Type of Source Clinker Cooler baghousesPermit No. None Is a Revised APEN Required? NoRemarks: NEDS PT ID : 01

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
PT	0.10 lb/ton feed to kiln	Reg No. 6 IV. C. 2a.	No emission factor listed SCC units: Tons of cement produced 454,296 ton/yr x 1 lb PT/ton x (1-0.999) baghouse x ton/2000 lb = <u>0.2 ton/yr</u>	3-05-006-14 latest EIS printout	yes
	10% opacity	IV C 2b	<10%	Direct Observation	YES

Martin Marietta Cement COM PANCE DETERMINATION WORK SHEET Source No. 3, 9/1/83

Emission Point 009 Type of Source Homogenizing silo

Permit No. Grandfathered Is a Revised APEN Required?

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	20% Opacity	Reg. No. 1 II. A. 1.	0%	Observation	yes

Emission Point 010 Type of Source Finished products silo

Permit No. Grandfathered Is a Revised APEN Required?

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	20% Opacity	Reg. No. 1 II. A. 1	0% 454,296 tons/yr x 0.24 lb/ft/ton x (1 - 0.999) baghouse x ton/2000 lb = <u>0.05 ton/yr</u>	Observation SCC Code 3-05-006-19	yes.

Martin Marietta Cement

COMPLIANCE DETERMINATION WORK SHEET

Source No. 3

9/1/83

Emission Point Φ11Type of Source Mill BuildingPermit No. Grandfathered

Is a Revised APEN Required? _____

Remarks: _____

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	20% Opacity	Reg. No. 1 II. A. 1.	0%	Observation	yes
			454,296 ton/yr x 32.0 lbPt/ton x (1-0.999) baghouse x (ton/2000lb) = 7.3 tons/yr	SCC Code p. 3.7.0-65 3-05-006-17	

Emission Point Φ12Type of Source Silica silosPermit No. Grandfathered

Is a Revised APEN Required? _____

Remarks: _____

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Pt.	20% Opacity	Reg. No. 1 II. A. 1.	Unknown.		